Project DT-II: Ontologies, Multi-agent Digital Twin and Intelligent Telemetry for the Internet Infrastructure

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Abstract

The Internet Infrastructure is dynamic and heterogeneous, which makes its governance difficult through manual procedures and vendor-specific models. This paper presents DT-II, an ontology-driven multi-agent Digital Twin designed to represent, monitor, and govern this infrastructure semantically. The proposal defines an Upper Ontology, grounded in the Unified Foundational Ontology (UFO) and modelled in OntoUML, centred on the triad `Node-Connection-Channel`. The model is formalised in Description Logic and operationalised as a Knowledge Graph, whose instance base (ABox) is continuously synchronised via model-driven telemetry (e.g., YANG-Push). On this basis, DT-II applies automated reasoning (TBox/RBox/ABox) to support semantic interoperability, fault detection/diagnosis, and compliance checks (e.g., SLAs and Ontological Compliance Regime). Finally, the system is organised as a multi-agent architecture based on MAPE-K, enabling explainable and policy-driven actions across heterogeneous network domains.

Keywords Domain Ontology · Foundational Ontology · UFO · OntoUML · NETCONF · Multi-agent Digital Twin

1 Introduction

The Internet Infrastructure (II) has become **highly** dynamic and heterogeneous, making it unreliable to govern solely through manual procedures, *ad hoc* scripts, and vendor-specific interpretations. In this context, ***Knowledge Graphs*** (KGs) have been employed to represent the operational state as connected facts, whilst ***ontologies*** provide the vocabulary,

constraints, and consistency criteria that render these facts interpretable and verifiable (Kejriwal, Knoblock, and Szekely 2021; Landgrebe and Smith 2025; Talisman 2026; Juliao Braga, Henriques, et al. 2025). In DT-II, the ontology defines the logical framework (classes, relations, and axioms), and the KG represents its instantiation, continuously updated.

This paper proposes a ***Upper Ontology***[1] (UO) for II (Arp, Smith, and Spear 2015), grounded in the Unified Foundational Ontology (UFO) and formalised in Description Logic (DL) (Baader et al. 2017; D. Nardi and Brachman 2010; Horrocks, Kutz, and Sattler 2006). In practical terms, the UO acts as a **Normative Ontological Basis**, which establishes core concepts and common rules to unify heterogeneous subdomains (e.g., 5G, satellite, and optical) around the triad **Node–Connection–Channel**. Modelling in OntoUML/OLED separates *types* bearing identity (Kind) from *functions* dependent on context (Role), which favours systematic validation and export to OWL/Turtle (Guerson et al. 2015; G. Guizzardi, Fonseca, et al. 2018; W3C OWL Working Group 2012). Complementarily, according to (D. Nardi and Brachman 2003), page 13, a typical DL-based knowledge base consists of two components: a ***TBox*** and an ***ABox***. They continue: the ***TBox*** contains intensional knowledge in the form of a terminology (hence the term "***TBox***"; but "taxonomy" could also be used) and is built through declarations that describe general properties of concepts. Due to the nature of subsumption relationships among the concepts that constitute the terminology, ***TBoxes*** are usually regarded as having a lattice-like structure; this mathematical structure is entailed by the subsumption relationship — it has nothing to do with any implementation. The ***ABox*** contains extensional knowledge — also called assertional knowledge (hence the term "***ABox***") — knowledge that is specific to the individuals of the domain of discourse. Intensional knowledge is generally considered not to change — "timeless", in a way — whereas extensional knowledge is seen as contingent, or dependent on a particular set of circumstances, and therefore subject to occasional or even constant change. (Baader et al. 2017), on page 209, describes that in DL $\mathcal{SROIQ}$, the $\mathcal{S}$ is used as an abbreviation of DL in its simplest form, named $\mathcal{ALC}$, and that the letter $\mathcal{R}$ denotes an extended set of axioms representing the roles, which constitute the so-called ***RBox***. Thus, there are three components that constitute a knowledge base in DL (or OWL): ***TBox***, ***ABox***, and ***RBox***, systematically characterised in this document and and shown in Figure 1.

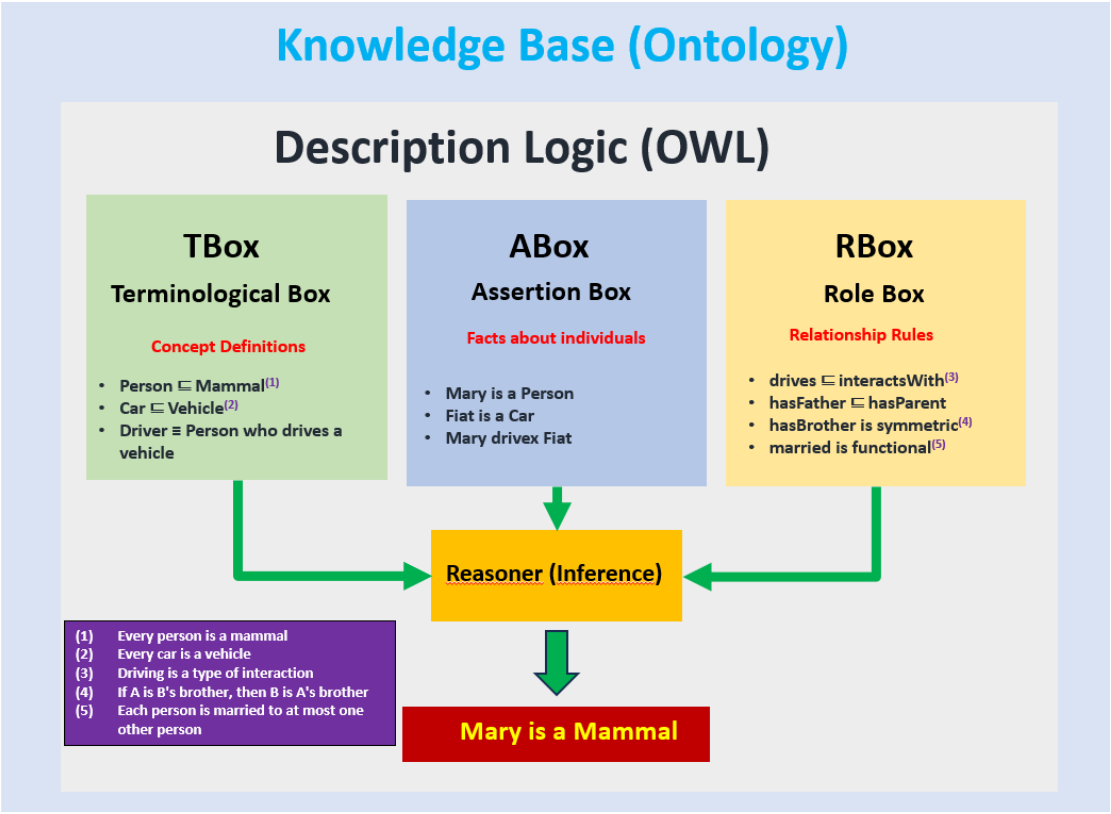


Figure 1: TBox, ABox e RBox form an ontology

[1]Domain-independent ontology that describes universal concepts applicable to any area of knowledge.

1.1 Types of Ontologies

There are several types of ontologies. This article proposes a ***Upper Ontology***[2] (UO) (Arp, Smith, and Spear 2015), grounded in the Unified Foundational Ontology (UFO), and a domain ontology, formalised in Description Logic (DL) (Baader et al. 2017; D. Nardi and Brachman 2010; Horrocks, Kutz, and Sattler 2006). In practical terms, the UO acts as an **Ontological Normative Basis**, which establishes core concepts and common rules to unify heterogeneous subdomains (for example, 5G, satellite, and fibre optics) around the triad **Node–Connection–Channel**.

UFO does not state ***what*** exists in the domain, but provides the categories and rules to correctly model what exists. In practice, it helps to answer questions such as "is this a type, a role, or a phase?" and "what gives identity to this entity?". It also prevents classic modelling errors, such as treating a role as a rigid subtype. It is usually employed via **OntoUML**, the modelling language based on UFO, and implemented in OWL through **gUFO**[3].

Applied to Internet Autonomous Systems, a sketch would be:

- **Organisation** (Kind): the entity that operates the AS.
- **Autonomous System**: an administrative entity identified by an ASN, not by a physical thing. In UFO, it may be treated as a social object, a Kind with identity given by the ASN and its assignment.
- **ASN / IP prefix assignment** (Relator): connects the Regional Internet Registry (RIR) to the organisation. It is the relator that ***creates*** the link.
- **Transit provider, Client, Peer** (Roles): no AS is essentially a provider. The AS plays this role within a relationship.
- **Transit agreement / Peering agreement** (Relators): underpin these roles. The social part of UFO models commitments and rights of the contract.
- **Active / Deactivated** (Phases): the AS changes phase without ceasing to be the same AS.
- **Route announcement, BGP session, prefix hijack** (Events): things that happen in time, i.e., they are not objects.

The main gain is the distinction between roles and relators. It models well the reality of commercial relations: the same AS may simultaneously be a client of one neighbour, a provider of another, and a peer of a third. A naïve taxonomy, with ***Transit AS*** as a subclass of AS, breaks precisely at this point.

UFO is divided into three main layers (G. Guizzardi, Falbo, and R. S. S. Guizzardi 2008; G. Guizzardi, Wagner, Falbo, et al. 2013), each built upon the previous:

- **UFO-A (endurants)**: things that ***exist as wholes*** at each moment and persist through time while changing properties. Includes objects, types (kinds, subkinds), roles, phases, qualities, modes, and relators. It is the most widely used layer and the basis of OntoUML. In this case, the organisation, the AS, and the ASN assignment are UFO-A entities.
- **UFO-B (perdurants / events)**: things that ***happen*** and unfold over time, existing in temporal parts. Covers events, their parts, participation, causality, and situations. A BGP route announcement, the establishment of a session, or a prefix hijack belong here.
- **UFO-C (social and intentional entities)**: agents, intentions, beliefs, goals, actions, commitments and claims, roles and social objects, norms. This is where contracts and peering and transit agreements are modelled, as well as the institutional assignment that creates the AS itself.

There are **later extensions**, built upon these three:

- **UFO-S (services)**: models service offerings, agreements, and deliveries between provider and client (J. C. Nardi et al. 2015). Highly relevant for transit relations.
- **UFO-L (legal)**[4]: refines UFO-C with legal positions in the style of Hohfeld and Alexy (right, duty, power, liability, immunity) (Griffo, Almeida, and G. Guizzardi 2018; Alexy 2002). Useful for contracts and regulation.
- **MLT / ML2 (multi-level)**: deals with types of types, i.e., when categories are instances of other categories (Almeida, Fonseca, and V. A. Carvalho 2017; V. A. d. Carvalho 2016; V. A. Carvalho and Almeida 2018). Example: ***AS type*** as a category whose instances are classes.

For an ontology of Autonomous Systems, the most natural set would be UFO-A + UFO-B + UFO-C. UFO-S is included if there is interest in modelling transit as a service, and UFO-L if agreements and RIR regulation are central.

[2] Domain-independent ontology that describes universal concepts applicable to any area of knowledge.

[3] `https://arxiv.org/abs/2603.20948`

[4] `https://nemo.inf.ufes.br/en/projetos/ufo-l/`

1.1.1 Advantages of using UFO in the development of domain ontologies

Some of the main advantages of using UO in constructing the domain ontology for II are described below:

V1. **Ontological distinctions that prevent common errors.** A person remains a person throughout their life, but is a student only while enrolled. UFO separates what something permanently is (type) from what is temporary or context-dependent (role or phase).
In the II domain: a network is an AS, but acts as a transit provider. Modelling ***Transit Provider*** as a fixed type of AS would be an error, since the same network may be a provider to one neighbour and a client to another simultaneously. (G. Guizzardi 2005a; Guarino and Welty 2002).

V2. **Explicit identity criteria.** A football team replaces all its players over the years and yet remains the same team. UFO requires one to state what guarantees this continuity.
In the II domain: an AS replaces routers, cables, and prefixes and yet remains the same AS. What sustains it is the official assignment by the registry (RIR), not the number nor the equipment. If an ASN is returned and reassigned to another company, a new AS is created with the same number. (G. Guizzardi, Benevides, et al. 2022).

V3. **Well-founded relations (relators).** A marriage is not merely a line connecting two people. It has a start date, rules, and may end. UFO treats such relations as entities in their own right.
In the II domain: a peering agreement has a start date, conditions (such as traffic exchange without charge), and may be altered or terminated. Modelled as a relator, the system can answer ***since when have these two networks been partners?***. (G. Guizzardi and Wagner 2008; Guarino and G. Guizzardi 2015).

V4. **Objects, events, and social aspects within the same framework.** To describe a football match, one must refer to things (players, ball), happenings (goals, fouls), and social rules (contracts, regulations). UFO has a layer for each, and they fit together.
In the II domain: the AS and the router are things (UFO-A). A BGP route announcement or a prefix hijack are happenings (UFO-B). The transit contract and the assignment by the RIR are social aspects (UFO-C). All are contained within a coherent single model (G. Guizzardi, Falbo, and R. S. S. Guizzardi 2008) (G. Guizzardi, Wagner, Falbo, et al. 2013).

V5. **Language and ready-made tools.** A spell checker points out errors as you write. OntoUML, the modelling language of UFO, does something similar with the model: it points out invalid constructions and suggests known patterns.
In the II domain: if one declares that "Client" is a type without stating from which type it inherits identity, the tool flags the error. It also suggests the correct pattern: Client is a role of an AS, mediated by a contract (G. Guizzardi, Wagner, Almeida, et al. 2015).

V6. **From reference model to implementation.** An architect first produces a complete and detailed plan, then the executive version for construction. UFO works in the same way: a rich model for people to understand and a simplified version (gUFO, in OWL) for computers to process.
In the II domain: the reference model documents with precision what an AS, an agreement, and an announcement are. The gUFO version feeds a queryable database, for instance to list all ASes that act as transit providers in Brazil. (G. Guizzardi 2007; Almeida, G. Guizzardi, Tiago Prince Sales, and Falbo 2019; Almeida, G. Guizzardi, Tiago Prince Sales, and Fonseca 2026).

V7. **Semantic interoperability.** If two cities use the same socket standard, any appliance works in both. Ontologies built upon UFO share the same conceptual "socket standard".
In the II domain: the AS ontology can be linked to UFO-S to treat transit as a service, to UFO-L for contracts, and to COVER to analyse the risk of a prefix hijack, without translating concepts from one to another (J. C. Nardi et al. 2015; Griffo, Almeida, and G. Guizzardi 2018; Tiago Prince Sales et al. 2018).

V8. **Maturity and validation.** No one builds a bridge with a method invented last week: one uses tested standards. UFO has more than two decades of use in diverse areas and is undergoing standardisation at ISO (ISO/IEC CD 21838-5).
In the II domain: when proposing an AS ontology to the networking community or in an article, grounding it in a mature foundational ontology in the process of standardisation provides credibility and reduces questioning of the method (G. Guizzardi, Benevides, et al. 2022; Almeida, G. Guizzardi, Tiago Prince Sales, and Fonseca 2026).

V9. **Justified modelling decisions.** A judge does not decide "just because": the ruling is grounded in law and jurisprudence. In UFO, each modelling choice has a theoretical basis that can be cited.
In the II domain: why is the AS distinct from the physical network? Because the network constitutes it without being identical to it, as marble to a statue. This explicit justification anticipates the question a reviewer would raise (Baker 2000; G. Guizzardi 2011),

UFO has a steeper learning curve, and the reference version is more expressive than OWL. Using it means higher initial costs, yet it prevents costly errors along the way. It is worthwhile when the domain is complex, involves relations and agreements, and when the model must endure and integrate with others, as is the case with Autonomous Systems.

The modelling of a domain ontology using the OntoUML tool, enabled by UFO, separates identity-bearing types (Kind) from context-dependent functions (Role), which favours systematic validation of the work being carried out and export to OWL/Turtle (Guerson et al. 2015; G. Guizzardi, Fonseca, et al. 2018; W3C OWL Working Group 2012).

1.2 Contributions

The main contributions of this work are:

- an ontology-driven reference model for II centred on *Node–Connection–Channel*, grounded in UFO and developed in OntoUML;
- an operationalisation in DL/OWL supporting automated reasoning over ***TBox***,***ABox***, and ***RBox***;
- a synchronisation approach aligning the KG with model-driven telemetry (e.g., YANG-Push) for near real-time state updates;
- a governance mechanism based on an **Ontological Compliance Regime** and SLA constraints, enabling auditable and explainable compliance checks; and
- a multi-agent Digital Twin architecture following the ***MAPE-K*** cycle (Kephart and Chess 2003; IBM Corporation 2005) for policy-driven actions.

1.3 Ontological Compliance Regime (overview)

The Ontological Compliance Regime (OCR) is a normative layer (in other words, a governance mechanism) defined by ontological axioms and mediated by relators (G. Guizzardi 2005b; Almeida, G. Guizzardi, Tiago P. Sales, et al. 2019; Almeida, G. Guizzardi, Tiago Prince Sales, and Fonseca 2026). It enables the Digital Twin to determine which constraints (e.g., legislation, SLAs, privacy policies) apply to entities and data flows, and to propagate these constraints across nodes, connections, and channels through automated reasoning.

The main characteristics and functions of this regime include:

- **Automated Propagation:** By employing a logical reasoner, the regime propagates normative constraints across nodes, connections, and network channels. If a node is bound to a specific legal norm, all connections and channels mediated by it inherit those same constraints.
- **Jurisdiction Redefinition:** In this model, jurisdiction ceases to be a mere static hardware attribute delimited by physical geography, and is instead understood as a normative entity based on meaning, context, and relations.
- **Legally Aware Infrastructure:** The regime transforms the infrastructure into a "legally aware" system, which means that the compliance of network operations is mathematically verified through DL.
- **Auditability and Explainability:** The regime ensures that compliance checks are both auditable and explainable. In dynamic events, such as failures or attacks, decisions taken by the MAPE-K cycle are only executed in the network after being logically validated by this normative base.

1.4 A practical roadmap for the DT-II project

DT-II is the name assigned to this project. This section describes its various components in detail.

The Figure 2 displays the proposed construction of the development architecture for the research project. It adopts an iterative and incremental cycle that articulates the Proof of Concept (PoC) and the Minimum Viable Product (MVP). While the Upper Ontology (TBox+RBox) provides rigorous conceptual grounding anchored in UFO/gUFO, the development of the Domain Ontology (ABox) occurs through continuous interaction between the emulated PoC and the MVP. The PoC acts as an agile testing environment for abstraction and modelling problem-solving in Containerlab[5], allowing the validation of conceptual competencies before deployment in the three operational complexity profiles of the MVP: ASN++ (high-complexity domain with global transit), ASN+ (medium-complexity edge/access domain), and IX (special traffic exchange domain). The entire ecosystem is maintained and reconfigured in near real-time by the Digital Twin through model-driven telemetry and automated reasoning.

[5]Containerlab is an open-source tool that enables the creation and management of container-based network laboratories in a fast, lightweight, and declarative manner, using YAML files to define topologies and interconnect real network operating systems.

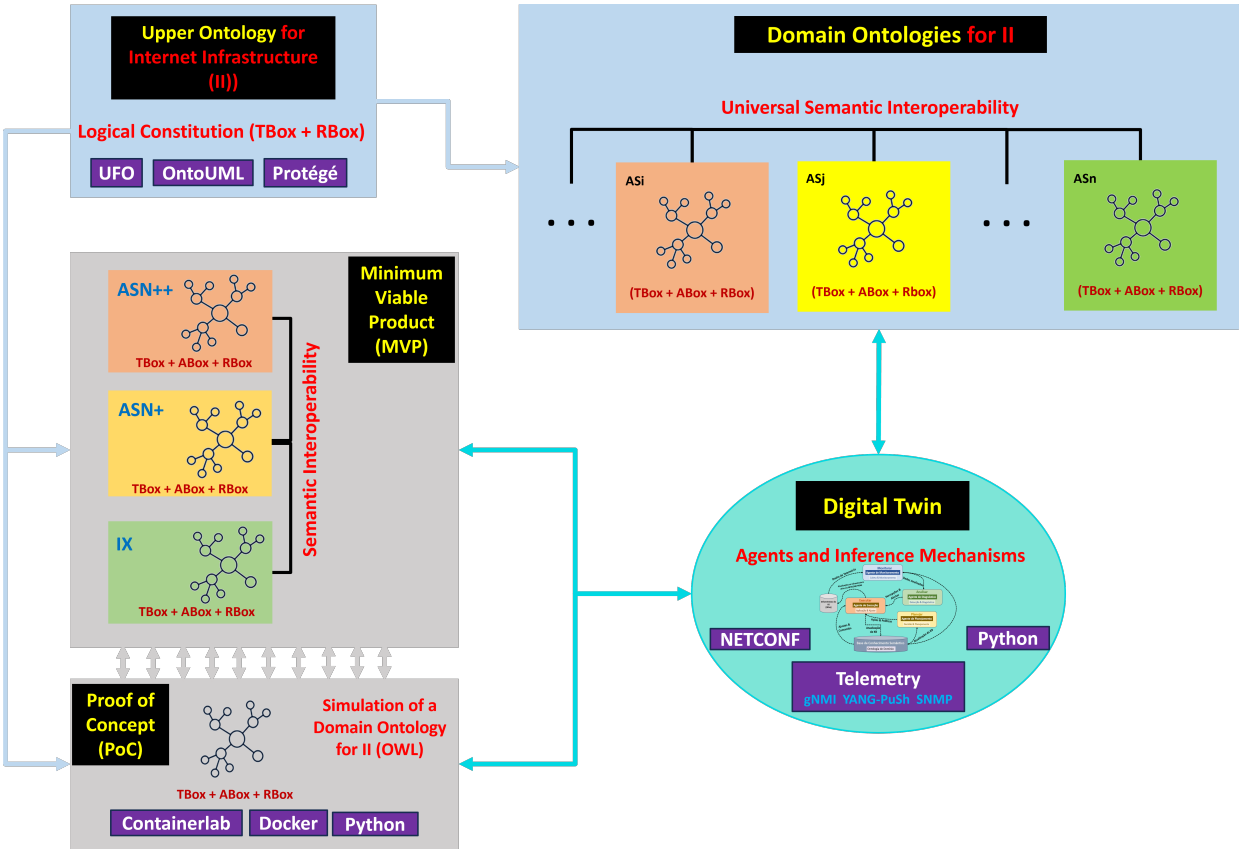


Figure 2: Architecture of the PoC and MVP proposal, across three operational environments with different characteristics in terms of complexity. These three types of complexity are classified as ASN++ (***High Complexity Domain***), ASN+ (***Medium Complexity Domain***), and IX (***Special Domain***), all characterised in the text that follows. There is continuous interaction between the PoC and the MVP, mainly aimed at solving problems during the development of domain ontologies: abstraction and problem-solving.

Table 1 characterises each of the complexities required for participants in the preliminary tests that will result in either the PoC or MVP. This is an initial approach to this classification and will be assessed *a posteriori* by other team members, with specific characteristics possibly being added or removed to ensure effective characterisation for the final choice.

The maintenance of the original proposal involving the construction of the PoC through simulation is strengthened by the following arguments, which together are methodologically sound:

- Speed in topology construction, i.e., reduced latency in learning: The PoC allows rapid iteration over modelling failures in OntoUML/Protégé and Python code without depending on complex topologies of multiple ASes.
- Heterogeneous representativeness: By extending the PoC to the MVP divided into ASN++, ASN+, and IX, the project gains strong validation arguments for the academic/IETF community, showing that the ontology is not limited to one type of network but unifies domains with entirely distinct operational and legal roles.
- Opportunistic change in project topology, without concerns related to clients, since in simulation they do not exist;
- Facilities in exercises and experiments that may lead to learning and improvement in the construction of domain ontologies, especially considering the perspective of topological abstraction, since we have UFO (or rather, gUFO). Topological abstraction can lead to a standard, implementable domain ontology with few changes in any Autonomous System[6] (AS). Additionally, it can assist in the gradual construction of the ABox, allowing small instances (synthetic ABox in the PoC) to be built and tested before dealing with the volume of assertions generated by continuous telemetry (gNMI, YANG-Push) across the three types of ASes in the MVP.

[6]An Autonomous System (AS) is the fundamental unit of routing on the Internet, defined as a set of IP networks under a single routing policy and administration.

Table 1: Classification of MVP participants, by level of complexity

High Complexity (ASN++)	Medium Complexity (ASN+)	Special Domain (IX)
Meet highly specialised and critical demands	Meet intermediate demands, with some level of specialisation	Meet a highly complex environment that centralises the interests of a large number of ASes of varying complexity
Robust infrastructure and advanced technology	Adequate infrastructure, but less sophisticated	Very complex infrastructure with highly diverse Internet traffic
Highly qualified multidisciplinary teams	Qualified technical teams, but less diverse	Highly qualified and diverse multidisciplinary teams
Services with high availability, redundancy, and security	Reliable services, but with less redundancy	Services with high availability and differentiated within the same environment
Integration with complex, large-scale systems	Moderate customisation capacity	Little or no integration capacity
Serve large corporations, public bodies, or strategic sectors	Serve medium-sized companies	Serve small, medium, large, and very large institutions with diverse service/-traffic demands
Bring the complexity of global routes, transit policies, and complex peering agreements	Bring the edge perspective, where regulations such as LGPD/SLA and local access policies apply	The neutral traffic exchange point, where the semantics of relators (multilateral BGP sessions, exchange agreements) are tested to the limit

- There is no urgency in this research proposal. Both options, PoC and MVP, can coexist in the project, and simulation will improve the characterisation of MVP participants and even provide evaluation.

The strategy of including the MVP through three environments involving different complexities favours the improvement of project learning and broadens the development and recording of knowledge and lessons learned.

On 2026-09-28, the Internet had approximately **79,473** active Autonomous Systems (ASs) (Hawkinson and Bates 1996), as reported in the **CIDR IPv6 Report**[7] maintained by Geoff Huston, Chief Scientist from Asia-Pacific Network Information Centre (APNIC). Each AS possesses operational autonomy, with its functionality established by protocols and specialised administrators. The correctness of this functionality characterises the behaviour of the Internet from the user's perspective. A misconfiguration in one AS can propagate across the global II, causing worldwide incidents. For example, if an AS incorrectly announces an IPv4 block via the BGP protocol, it may cause serious damage, particularly to the legitimate holder of that block. One way to prevent anomalous behaviours in the AS environment is through measurements known as **telemetries**.

These risks highlight the need for advanced monitoring mechanisms. The DT-II project proposes the simulation of ontology-based techniques (or knowledge bases) capable of establishing intelligent telemetries without human intervention. Alongside ontologies, protocols such as NETCONF and related technologies are employed, widely deployed and implemented by major II equipment vendors.

The project employs two types of ontologies: domain ontologies (DOs), which specify the knowledge of each AS, and the UO, which ensures semantic interoperability and facilitates the construction and verification of DOs through the OntoUML tool. Although DOs are similar across most ASs, the innovation of the project lies in the use of the UO, which brings benefits such as:

- Ensuring semantic interoperability across different domains;
- Facilitating systematic construction and verification of DOs via OntoUML.

Additionally, the project uses *Protégé*, supported by gUFO, for modelling DOs both in the testbed and in ASs participating in the second phase of the research. The testbed, developed with ***Containerlab***, ***Docker***, and ***Python***, constitutes the DT-II PoC, enabling validation of the project's implementation.

The DT is the operational component of the project. It adopts the autonomic characteristics described in Richard Murch's book "Autonomic Computing" (Murch 2004) and is expected to behave in accordance with the eight elements proposed to preserve the characteristics of an autonomic computing system:

- Comprehensive self-knowledge, including available resources and operational capabilities
- Ability to configure and reconfigure itself autonomously in response to contextual requirements

[7] `https://www.cidr-report.org/`

- Capacity for continuous self-optimisation, ensuring sustained efficiency and performance
- Robust self-healing mechanisms to detect, isolate, and recover from failures
- Self-protection capabilities to safeguard against internal inconsistencies and external threats
- Ability to acquire and interpret knowledge of its environment and context, adapting behaviour accordingly
- Competence to operate effectively in heterogeneous and distributed computing environments
- Ability to anticipate, interpret, and adapt to evolving user requirements

A DT can maintain an environment that preserves the above elements (e.g., self-knowledge, self-configuration, self-optimisation, self-healing, and self-protection) if conceived as a **multi-agent** architecture supported by a semantic knowledge base. To this end, the DT follows the **MAPE-K** proposal (Kephart and Chess 2003; IBM Corporation 2005), which establishes the following cycle as a means of ensuring autonomy:

- **Monitoring (M)**: collects real-time data.
- **Analysis (A)**: interprets the data based on axioms and rules.
- **Planning (P)**: generates action plans consistent with the Normative Ontological Basis.
- **Execution (E)**: applies changes to the infrastructure.
- **Knowledge (K)**: feeds back into the knowledge base, ensuring continuous learning.

Figure 3 provides a detailed visual representation of the MAPE-K flow of the DT proposed in this article.

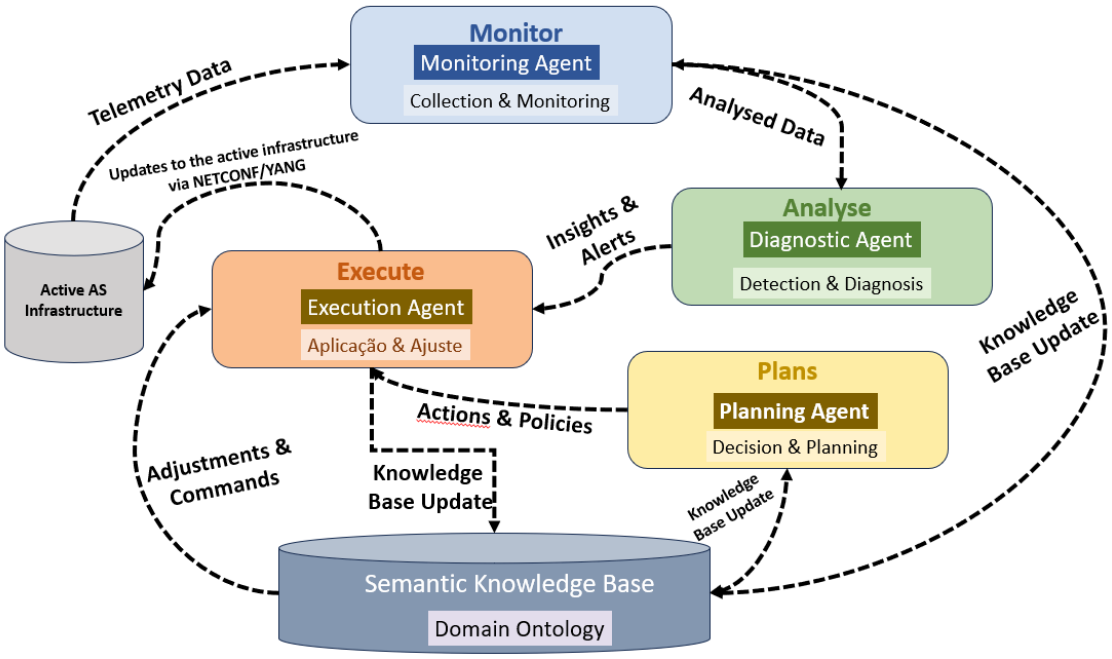


Figure 3: MAPE-K Flow of the Digital Twin

2 Related Work

This section positions DT-II in relation to three research strands: (i) **network digital twins** and closed-loop/autonomic management; (ii) **ontology-based network modelling** and knowledge graphs for inventory, topology, and intent; (iii) **telemetry-driven inference** and policy/compliance enforcement.

2.1 Network Digital Twins and Autonomic Control

Digital Twins provide a digital counterpart of an operational system, enabling monitoring, diagnosis, and actuation (Grieves and Vickers 2017). In networking, recent proposals emphasise closed-loop control, model-driven operations, and integration with telemetry (Zhou et al. 2026). DT-II differs by making the twin *explicitly governed* by a superior Normative Ontological Basis, so that actions are validated against semantics and constraints rather than solely metrics or learned policies.

2.2 Ontologies and Knowledge Graphs for Network Infrastructure

Knowledge graphs have been applied to integrate heterogeneous network data sources and to support inter-domain queries such as in Thomas et al. (2026). Ontology-driven approaches add explicit meaning, identity criteria, and constraints, enabling consistency checking and explainable inference. DT-II builds upon this line by grounding its central model in UFO (via OntoUML/gUFO), explicitly modelling roles and relators, and using this structure as a foundation for interoperability across multiple administrative domains.

2.3 Telemetry, Model-driven Interfaces, and Reasoning

Modern network management increasingly relies on streaming telemetry and model-driven interfaces (e.g., YANG-based mechanisms) to maintain near real-time state. DT-II adopts model-driven telemetry via YANG-Push (RFC 8641) (Clemm and Voit 2019), as its synchronisation mechanism, combining it with DL reasoning over ***TBox/RBox/ABox*** to detect inconsistencies and to derive governance-relevant states (e.g., SLA violations, jurisdictional conflicts).

2.4 Governance, Policies, and Compliance

Whilst policy frameworks typically impose restrictions through configuration templates or procedural checks, DT-II formalises governance as part of the ontology itself (e.g., Ontological Compliance Regime, SLA relators). This enables automated validation, traceability, and explainability of compliance decisions across domains.

3 Methodology

This project follows the ontology engineering lifecycle (Keet 2025; Parks 2025; Effingham 2013), employing the ***Unified Foundational Ontology*** (UFO) as its basis and OntoUML as the modelling language (G. Guizzardi 2005b). The process is structured into four technical phases:

M1. **Conceptual abstraction**: UFO is applied to classify II assets. Rigid essence types (`PhysicalElement`, `MaterialMedium`) provide identity, whilst anti-rigid roles (`NetworkNode`, `CommunicationChannel`) are mediated by relators.

M2. **Formalisation in Description Logic (DL)**: The resulting conceptual model is axiomatised in DL to ensure computational interpretability. For example, every `NetworkNode` must possess a logical identifier:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement} \sqcap \exists \text{possesses.LogicalIdentifier} \tag{1}$$

M3. **Semantic serialisation**: Models validated in ***OLED*** are exported to OWL and serialised in ***Turtle***. Standard vocabularies (e.g., ***Dublin Core*** and ***FOAF***) provide support for metadata and provenance.

M4. **Automated reasoning**: DL reasoners query the ***TBox*** and ***ABox*** to check consistency and to derive governance-relevant states. For instance, an inactive channel can be used to infer compromise (or invalidation) of the corresponding logical connection.

3.1 Ontological Analysis Structure and Mapping Process

The mapping between the II domain and UFO, via its gUFO implementation, follows a specialisation protocol grounded in logical metaproperties. For each local domain entity, the **Ontological Analysis Structure** is applied (Almeida, G. Guizzardi, Tiago Prince Sales, and Fonseca 2026; Almeida, G. Guizzardi, Tiago P. Sales, et al. 2019; G. Guizzardi 2005b):

A. **Specialisation of Fundamental Types**: Entities with rigid identity and specific individuation principles (e.g., Physical Routers, Satellites, Fibre Cables) are mapped as subclasses of *gufo:Kind*. These represent substantial objects that persist independently of their configuration state.

B. **Relational Specialisation (Roles)**: Concepts whose existence depends on a specific topological or functional context (e.g., Edge Node, Peering Point, Transit AS) are characterised as *gufo:Role*. This allows the Digital Twin to capture the dynamic nature of network functions independently of the underlying hardware.

C. **Relational Foundations**: Logical connections and service agreements establishing dependencies between multiple objects (e.g., BGP Sessions, SLA Agreements) are formalised as *gufo:Relator*. This ensures that the integrity of the Knowledge Graph is maintained through existential dependency constraints inherent to gUFO, where invalidation of a relator triggers logical removal of the associated roles.

By employing this structure, DT-II transcends mere data modelling, establishing a **Normative Ontological Basis** in which network behaviour is governed by foundational ontological axioms rather than purely heuristic or statistical patterns.

3.2 Dynamic Synchronisation via YANG-Push

A critical requirement for DT-II is high-fidelity synchronisation between the physical infrastructure and its semantic representation. To achieve this, the model employs *Model-Driven Telemetry* (MDT) through the **YANG-Push** mechanism (RFC 8641 (Clemm and Voit 2019)).

Unlike traditional query-based monitoring (*pull*), YANG-Push enables Network Nodes to directly transmit state changes to the *Semantic Ingestor*. This integration is mapped into the gUFO framework as follows:

- **Periodic Subscriptions**: Quantitative metrics, such as interface counters, are transmitted to update `gufo:Quality` assertions in the ABox.
- **Change Notifications**: Critical events, such as oscillations in BGP sessions, are treated as `gufo:Events` that immediately trigger re-evaluation of existential dependencies in `gufo:Relators`.

This ensures that the Normative Ontological Basis of the Digital Twin always reflects the current operational state of ASs with latency below one second.

4 Relations between Ontology, Knowledge Graph, and Description Logic

The ***Knowledge Graph*** (KG) operationalises the ontology by linking **real data** (instances) to the conceptual model. It represents information as **nodes** (entities) connected by **edges** (relations). For example, an instance `inst:Router_R1` may be linked to `inst:Router_R2` via `ii:connectedTo`, and the resulting `ii:Connection` may be linked to an `ii:Channel` with capacity and latency qualities. Typically, a KG integrates multiple sources to form a network of interconnected facts.

Ontology and KG are **complementary**: the ontology provides semantics and logical rules, whilst the KG instantiates them as a dynamic and queryable knowledge base. Table 2 summarises their roles in the proposed framework.

Table 2: Complementarity between Ontology and Knowledge Graph

Characteristic	Ontology	Knowledge Graph
Focus	Definitions, rules, logic.	Data, facts, connections.
Level	Abstract (the "DNA").	Concrete (the "Organism").
Function	Provides meaning and semantics.	Enables queries and discoveries.

Description Logic (DL) underpins modern ontologies (e.g., OWL in the Semantic Web) (Baader et al. 2017). DL enables the formal definition of concepts and computational reasoning. For example, the axiom:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement}$$

establishes that every `NetworkNode` is a `PhysicalElement`. More complex definitions employ logical connectives, such as:

$$\text{NetworkNode} \sqsubseteq \exists \text{possesses.LogicalIdentifier}$$

indicating that every `NetworkNode` possesses at least one `LogicalIdentifier`. Here, $\sqcap$ denotes intersection (AND) and $\exists$ denotes existence, enabling precise expression of semantic constraints.

In summary, the ontology provides the logical and semantic structure, whilst the KG materialises this structure in concrete data. DL acts as the formal link between them, ensuring consistency, interpretability, and inferential capability. This triad — Ontology, KG, and DL — constitutes the methodological core of DT-II, enabling the II to be represented, monitored, and governed in a semantic and dynamic manner.

5 Conceptual Structure of the Digital Twin

This section presents a **reference architecture** for implementing the II Digital Twin as an **operational KG** (***ABox***) governed by an ***UO*** (***TBox***, ***RBox***). The aim is to make the model executable: to collect, normalise, validate, infer, and support decisions. The implementation objectives are:

(a) **Unified semantic inventory**: represent nodes, connections, and channels independently of technology;
(b) **Continuous synchronisation**: align the KG with telemetry and topology changes;
(c) **Detection and diagnosis**: identify inconsistencies, failures, and cascading impacts;
(d) **Verifiable governance**: encode and audit rules (SLA, jurisdiction, authorisation).

5.1 Minimal Data Model

The initial implementation focuses on three entities:

(a) **Node**: identity (`hasIdentifier`), operational state, timestamp;
(b) **Connection**: endpoints (`connectNode`), supported channel, metrics (latency, loss, jitter);
(c) **Channel**: medium type (fibre, copper, radio), capacity, QoS policy.

5.2 Integration of *ABox* and *RBox*

The ***ABox*** instantiates infrastructure elements (routers, links, channels) from inventory data (Claise, Clarke, and Lindblad 2019). The ***RBox***, based on $\mathcal{SROIQ}$ logic (Horrocks, Kutz, and Sattler 2006), governs relational properties:

(a) **Transitivity**: deduces cascading failures;
(b) **Mereology**: validates part–whole relations for SLA compliance (Effingham 2013, Ch. 8);
(c) **Property chains**: enable bidirectional navigation and anomaly tracing.

Together, ***ABox*** and ***RBox*** transform the KG into a computable model, enabling dynamic reasoning and governance.

5.3 Operational Flow

F1. Ingestion → normalisation → triple generation.
F2. Validation → integrity checks and anomaly detection.
F3. Inference → classification and rule-based derivations.
F4. Query/alert → recommendations or automated orchestration.

5.4 Example Scenario

In a link failure:

(a) telemetry marks the `Channel` as unavailable;
(b) inference classifies the `Connection` as `InactiveConnection` and derives `SLAViolation`;
(c) redundancy is triggered (e.g., satellite backup).

In summary, the Digital Twin evolves into an **executable knowledge system**, where the ***TBox*** governs meanings and rules, and ***ABox/RBox*** capture state, validate consistency, and enable inference, transforming data into actionable decisions.

5.5 Reference Architecture

As a consequence, the reference architecture can be established, organised into seven layers:

(i) **Ontological Layer (*TBox*)**: defines core classes (`NetworkNode`, `Connection`, `Channel`) and constraints;
(ii) **Identity Layer**: assigns stable IRIs to instances, separating identity from state;
(iii) **Ingestion Layer**: collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalises data into the Node–Connection–Channel triad;
(iv) **Persistence Layer (*ABox*, *RBox*)**: stores facts in RDF/OWL and maintains temporal history via observations;
(v) **Validation Layer**: applies structural constraints and logical consistency checks with DL reasoners;
(vi) **Reasoning Layer**: infers states (`DegradedConnection`, `SLAViolation`) and propagates failures;
(vii) **Exposure Layer**: provides SPARQL queries, dashboards, alerts, and integration with NOC/orchestration systems.

A graphical view of these seven layers is available in Figure 4.

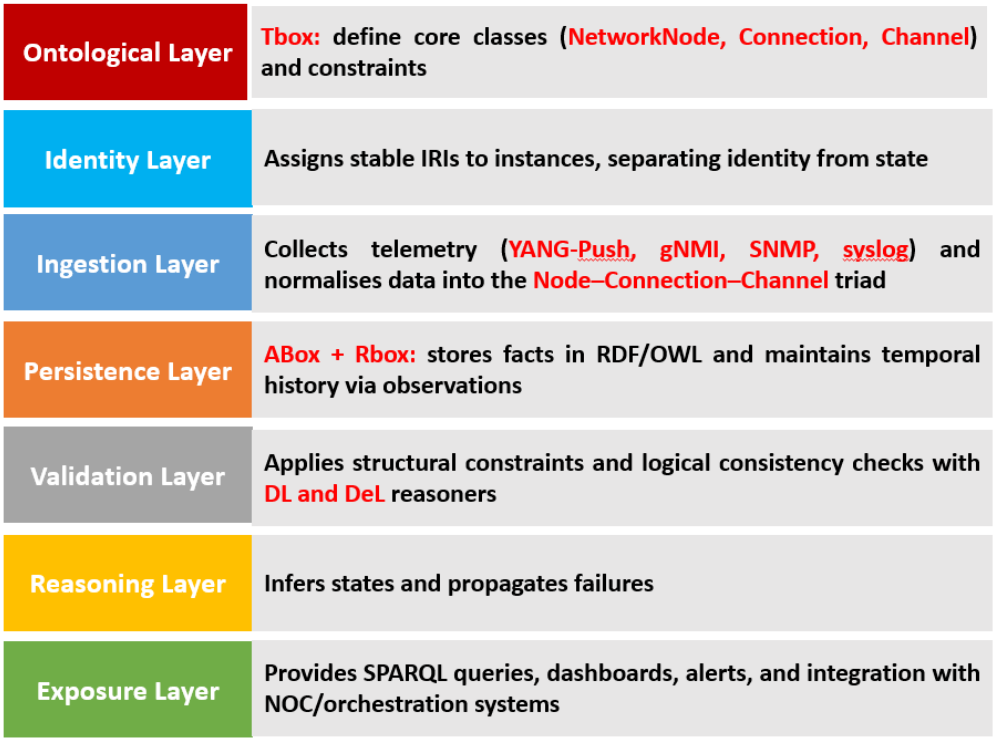


Figure 4: Graphical view of the Reference Architecture of the DT Conceptual Framework

6 Use Cases

6.1 Use Case I: Meaning-Driven Regulatory Compliance

In traditional networks, the law that applies to a data flow depends on where the equipment is physically located. In DT-II, the question changes: *what data is this, who is involved, and which agreements bind them?* The **OCR** (characterised in Section 1.3) is the governance mechanism that answers this question. Under the OCR, jurisdiction ceases to be a fixed attribute of hardware and becomes a **normative relationship**. This relationship is modelled as a *relator*, that is, an entity that connects a norm to the parties and network resources to which it applies.

Illustrative scenario

A hospital in São Paulo sends patients' test results to an analysis service in Europe. The flow is subject to both Brazil's General Data Protection Law (LGPD) and the GDPR, and it traverses three Autonomous Systems: AS-A (Brazil), AS-B (international transit) and AS-C (Europe). The rule to be upheld is that health data may only traverse nodes under jurisdictions that offer an adequate level of protection.

6.1.1 How the Rules Are Represented

The DT-II knowledge base comprises three complementary parts:

- **TBox — the general rules.** It defines the concepts and the constraints that always hold. Example: *'every flow carrying health data may only traverse nodes under a jurisdiction with adequate protection'*.
- **RBox — how relationships propagate.** It defines chains of relations (*property chains*). Example: *if a node is subject to the LGPD, then the channels and connections that pass through it are subject to it as well*. Thus, the obligation need not be declared resource by resource.
- **ABox — the facts of the moment.** It describes the current state of the network. Examples: *'flow F carries health data'*, *'F passes through node N_3'*, *'N_3 is under jurisdiction J'*.

The **reasoner**, a logical inference engine, combines rules and facts to infer automatically which norms apply to each flow and whether any of them is being violated.

6.1.2 The Role of the Digital Twin

The Digital Twin acts as a **continuous auditor** of the network. It collects telemetry via gNMI, YANG-Push and SNMP, which arrives as raw data (interfaces, routes, counters). Through **Semantic Elevation**, these data are translated into meaningful facts and inserted into the ABox, such as *'flow F, carrying health data, passes through node N_3'*.

In the scenario, suppose the telemetry indicates a route change: flow F now traverses a node in a jurisdiction without adequate protection. The reasoner detects that this new fact contradicts a TBox rule and flags the violation.

6.1.3 Automated Response: the MAPE-K Loop

To respond to dynamic events, such as a DDoS attack or a BGP failure that diverts routes, the OCR steers the **MAPE-K** loop. This loop originates in autonomic computing and comprises four stages that share a common knowledge base (*Knowledge*), which in DT-II is the ontology itself. Table 3 shows the loop applied to the scenario.

Table 3: The MAPE-K loop applied to the illustrative scenario.

Stage	What happens	In the scenario
Monitor	Telemetry updates the ABox	F is found to have changed route
Analyse	The reasoner checks the rules	F is found to traverse a non-compliant jurisdiction
Plan	Alternatives that satisfy the rules are sought	A route via AS-D, also compliant, is identified
Execute	The configuration is applied via NETCONF, *only after logical validation*	The new route is activated and F is compliant again

Summary

The OCR makes the infrastructure *jurisdiction-aware*: the network 'knows' which rules apply to each flow and verifies compliance through logical inference (Description Logic), rather than relying on manual checks. This preserves governance even in complex and constantly changing networks.

6.2 Use Case II: Responding to a DDoS Attack

In a Distributed Denial-of-Service (DDoS) attack, a large number of compromised machines flood the network with traffic until links and services run out of capacity. In this scenario, response time is critical. Conventional defences react to the *symptom*: they identify suspicious IP addresses and block them. DT-II reacts to the *meaning* of the event: it knows which services depend on the congested link, which of them are critical, and which agreements bind the network to the source of the attack. To this end, the Digital Twin acts as a **semantic defence system**, combining the knowledge base (***TBox***, ***RBox*** and ***ABox***) with the ***MAPE-K*** loop introduced in Use Case I.

Illustrative scenario

An external *botnet* attacks the network. Traffic entering through the edge link `Link_01` jumps from 1 Gbps to 100 Gbps and saturates the link. This link carries the traffic of the DNS service, which is deemed critical: if DNS stops responding, virtually every other service becomes unreachable for users.

6.2.1 The Four Stages of the Response

Monitor

Telemetry (gNMI, YANG-Push, SNMP) reaches the Digital Twin, which converts it, through Semantic Elevation, into facts in the ***ABox***:

```
inst:Unusual_Flow a ii:AnomalousFlow    inst:Link_01 a ii:Saturated
```

In plain terms: *'there is an anomalous flow'* and *'link `Link_01` is saturated'*.

Analyse

The reasoner combines these facts with the propagation rules in the ***RBox***. One of them is a property chain:

$$saturates \circ supports \sqsubseteq threatens$$

That is, *if a flow saturates a link and that link supports a service, then the flow threatens that service*. Since the dependency relation between services is transitive, the risk also propagates to the services that depend on DNS. The reasoner concludes, without anyone having declared it:

```
inst:Service_DNS a ii:AtRisk
```

It also identifies the neighbouring AS through which the traffic enters and the agreement (relator) in force with that AS, under the Ontological Compliance Regime described in Use Case I.

Plan

By querying the ***TBox***, the system applies a rule such as: *'if a critical service is at risk and the threat originates externally, mitigation must take place at the network edge'*. The options are *blackholing* (discarding the traffic destined for the target of the attack) and *rate limiting* (capping the rate of accepted traffic). The chosen plan prioritises the preservation of critical services and is validated against the rules of the Ontological Compliance Regime before it is executed.

Execute

The logical decision is translated into technical commands: via ***NETCONF***, the edge routers are reconfigured to filter the malicious traffic. The router confirms the change and the Digital Twin updates the ***ABox*** (state: *protected*). This closes the loop: at the next monitoring cycle, the system checks whether the attack has in fact been contained.

6.2.2 Sequence Flow

Figure 5 shows the complete loop, from the **Active Infrastructure** to the **Logical Core** and back, with no human intervention. Table 4 maps each ***MAPE-K*** stage to the numbered steps in the figure.

6.2.3 Advantages of the Semantic Approach

Compared with defences that merely block IP addresses, DT-II offers four advantages:

- **Context awareness.** If the attack originates from an AS with which a cooperation agreement exists, represented as a relator in the ontology, the system can engage that partner through governance APIs, for instance by asking it to filter the traffic at source, rather than simply blocking it.

Table 4: ***MAPE-K*** stages and the steps in Figure 5.

Stage	Steps	What happens
Monitor	1–2	Telemetry reveals the saturation and the fact is declared in the ***ABox***
Analyse	3–4	The reasoner infers that DNS is at risk
Plan	5	The knowledge base deduces the mitigation (*blackholing*)
Execute	6–8	The filter is applied via ***NETCONF***, confirmed and recorded in the ***ABox***

- **Mereological resilience.** The ontology knows which parts make up the network and the function of each (*mereology* is the theory of part–whole relations). The reasoner can therefore sacrifice non-vital channels to preserve critical services.
- **Traceability.** Every mitigation action can be explained by the chain of ***TBox*** and ***RBox*** axioms that justified it, which supports auditing and the operator's trust in the automation.
- **Compliance.** No configuration is applied without first being validated against the rules of the Ontological Compliance Regime, integrating defence with the governance described in Use Case I.

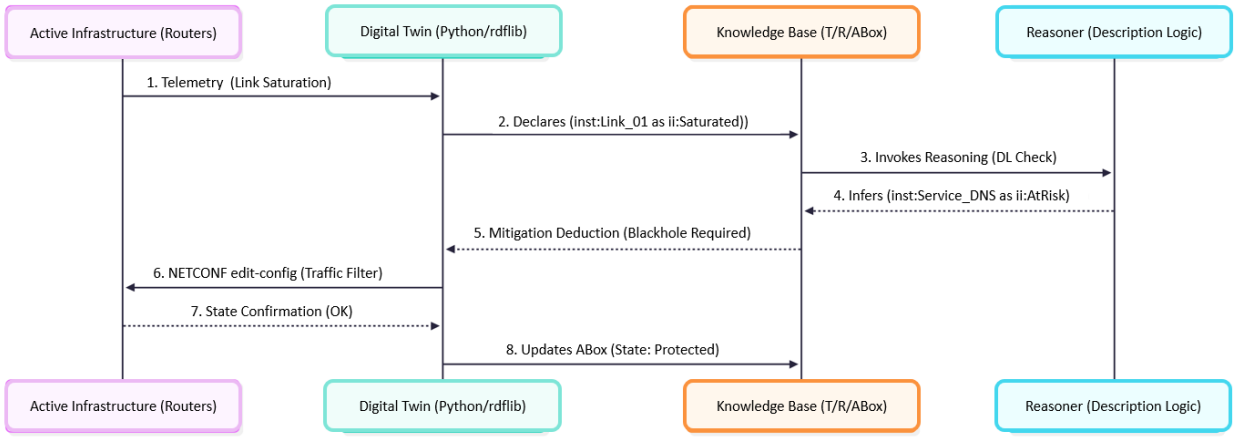


Figure 5: Sequence Diagram of DDoS Mitigation via DT-II

In summary, the ontology-based approach enables **policy-driven mitigation**, where assets of higher semantic and contractual value are prioritised, ensuring resilience and governance integrity.

7 Why the II Requires a Project like DT-II

The II must be understood not merely as cables and protocols, but as a **living system** that has already surpassed the limits of manual management. The **DT-II** project is necessary for four reasons: (R1) **From Static Management to Semantic Autonomy**: Current II management relies on rigid scripts and manual CLI commands. DT-II introduces **Logical Reasoning**, enabling the system to deduce solutions for complex failures and attacks under an ***Ontological Normative Base***; (R2) **NETCONF as an Interoperability Catalyst**: The growing adoption of ***NETCONF*** provides a vendor-independent, standardised channel for the Digital Twin to interact with the **Active Infrastructure**, updating ***ABox*** instances in real time; (R3) **Global Semantic Governance**: In a fragmented regulatory scenario (e.g., LGPD, GDPR), DT-II encodes laws and SLAs as axioms, ensuring compliance regardless of geographic routing complexity; (R4) **Universal Semantic Interoperability**: Grounded in the ***UO*** (that is, ***UFO***), DT-II resolves data silos, acting as a universal translator across domains (5G, Satellite, SD-WAN[8]), reducing costs and integration errors.

In summary, without DT-II, II remains an **Active Infrastructure** — processing packets without context. With DT-II, II gains **State Awareness**, evolving into an **Autonomic Network** that understands what it moves, why it moves, and whether it has logical permission to do so.

[8]Software-Defined Wide Area Network, a software-based approach to WAN management.

8 Conceptual Expansions of DT-II

This section deepens theoretical and practical aspects that complement the initial DT-II proposal, consolidating foundations and anticipating implementation challenges.

8.1 Inter-Domain Multi-Agent Coordination

Although the MAPE-K architecture is already described for internal agents, it is necessary to detail how multiple *Digital Twins* from different ASs interact. Inter-agent communication protocols such as FIPA-ACL or KQML may be employed for message exchange between graphs. In distributed attack scenarios (e.g., DDoS), planning agents from different ASs must negotiate routes and mitigation strategies, establishing consensus or federation among multiple MAPE-K cycles (Weyns et al. 2012).

8.2 Computational Scalability and *Stream Reasoning*

The use of expressive description logic (SROIQ) ensures semantic rigour but may generate bottlenecks in globally scaled ABoxes. To maintain millisecond-level response times, we propose the use of *Stream Reasoning* techniques and logical partitions of the ABox. This approach enables incremental reasoning over continuous data streams, ensuring DT-II operates in critical environments without compromising performance (Quin, Weyns, and Gheibi 2021; Andersson et al. 2023).

8.3 Conflict Resolution in the Ontological Normative Base

The Ontological Compliance Regime evaluates plans against strict rules, yet conflicts are inevitable in real infrastructures. For instance, the self-protection rule may clash with high-availability SLA requirements. Indeed, dealing with conflicting decisions and the adaptation actions they imply is recognised in the literature as one of the main open challenges for advancing self-adaptive systems (Quin, Weyns, and Gheibi 2021). To address this limitation and handle such situations, we suggest the use of defeasible logic[9] or ontological weight hierarchies (Antoniou et al. 2001; Nute 2003; Governatori and Maher 2017; Governatori, Olivieri, et al. 2013; Governatori, Padmanabhan, et al. 2009; García and Simari 2004; Grimm and Hitzler 2008; Lutz, Manière, and Nolte 2023; Sengupta, Krisnadhi, and Hitzler 2011). This approach enables the inference engine to resolve conflicting axioms in a consistent and transparent manner, within non-monotonic reasoning.

9 Conclusion and Future Work

This paper presented the conceptual foundations of **DT-II**, a multi-agent Digital Twin for II based on Ontology, Knowledge Graphs, and Description Logic. By formalising governance through the **Ontological Compliance Regime** and enabling policy-driven responses to complex events such as DDoS attacks, the model advances the vision of transforming II from a passive infrastructure into a **juridically aware autonomic system**. The proposed architecture demonstrates how ontological rigour can ensure resilience, compliance, and semantic interoperability in heterogeneous domains. All project results will be available at `https://github.com/LJuliaoBraga/dtii`.

Future work will focus on:

(i) simulation of the Digital Twin to evaluate alarms in ASs;

(ii) experimental validation of spatio-temporal response mechanisms;

(iii) integration with programmable infrastructures via ***NETCONF***;

(iv) exploration of semantic interoperability in domains such as 5G, satellite, and SD-WAN;

(v) participation in the evaluation of the two proposed telemetry alternatives — YANG-Push and gNMI — while remaining open to those that may emerge in the near future; and

(vi) Integration with Quantum Key Distribution (QKD) and Quantum Internet: We plan to explore the use of Quantum Communication channels for transmitting highly sensitive telemetry and governance metadata between the Active Infrastructure and the Digital Twin. As DT-II deals with critical "Semantic Jurisdictions" involving

[9] It is a logical system that admits conclusions based on rules that may be defeated by other rules or contrary evidence. In classical logic, once something is deduced, it cannot be withdrawn (monotonic reasoning). In defeasible logic, however, conclusions may be revised.

state-level privacy and strategic infrastructure topology, QKD integration will ensure synchronisation between the physical and logical layers remains secure even against future adversarial threats from quantum computing, maintaining the integrity of the "Ontological Normative Base" through a proven secure quantum-classical hybrid architecture (Kozlowski et al. 2023; Wang et al. 2024; Meter 2014). Future integration with Quantum Key Distribution (QKD) and Quantum Internet is a strategic differentiator of DT-II. We propose a hybrid architecture in which the Ontological Normative Base connects to quantum hardware to guarantee inviolable channels between the active infrastructure and the *Digital Twin*. This layer ensures that the MAPE-K cycle is not compromised by adversarial injections or tampering, reinforcing system resilience and security. By incorporating inter-domain multi-agent coordination, scalability via continuous synchronisation, normative conflict resolution, and quantum hybrid architecture, DT-II evolves from a conceptual proposal into an applicable engineering *framework*, capable of addressing the practical challenges of the Internet of the future.

In addition to the proposals listed above, it is worth highlighting the possibility of applying to DT-II the intelligent agents' architecture developed by Juliao Braga (2019) and previously explored in Julião Braga, Omar, and Granville (2015), and later in Juliao Braga, Stiubiener, and Henriques (2025). This integration may enhance the capacity for knowledge acquisition in restricted domains and strengthen the governance of ASs, contributing to greater efficiency, security, and explainability of the indicated actions, particularly in the application of the MAPE-K flow (Figure 3).

These steps aim to evolve the conceptual model design towards experimental deployment, culminating in its submission in 2026 as a draft to the IRTF ***nmrg*** within the CGI-IETF initiative, in collaboration with the following Brazilian universities: **IFSPe** (PE), **ITA** (SP), **UFABC** (SP), **UFRGS** (RS), and **UNICAP** (PE).

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Declaration on Generative AI

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